

1 User-Agent Interaction

Session 1 - Session 2 - Session 3

S1 (earlier)

Recommend a nearby Western restaurants for me. I like it.



I'll recommend Western food for you first. Do you like steak?



S2 (current conversation)

I tried Sichuan restaurant today. So delicious! I love spicy food, my frist choose.



Sounds good. I'll prioritize recommending spicy food.



S3 (later)

I'm upset. I've got gastroenteritis and have to eat light food recently.



Sorry to hear that. It must be tough. Rest well



2 Capture

Extract memories from current session (S2). Denoted as C_2^* .

user_profile

This user loves spicy food.

agent_method

Recommend spicy food frist.

3 Update

Incorporate C_2^* with prior memory bank B_1^* .

DELETE (from B_1^*)

This user likes Western food.

ADD (from C_2^*)

This user likes spicy and Western food.

Incorporate C_3^* with prior memory bank B_2^* .

DELETE (from B_2^*)

Recommend spicy food frist.

ADD (from C_3^*)

Recommend light and no spicy food.

Prior gold terminal memory, denoted as B_{T-1}^*

4 DyadMem Memory

After Session 3, denoted as B_3^* .

Short-term

episodic

User is sick on September 26.

User-side

user_profile & fact

user_profile

User is sick and needs light food.

URAM

agent_method

Recommend light and no spicy food.

User-conditioned

Other Memory

Cross user, General Workflow, Reusable task strategy.

User-side

user_profile & fact

user_profile

User is sick and needs light food.

General Agent Memory

agent_method

Recommend spicy food frist.

5 Recall

Use relevant memories to generate the responses.

Query

What can you recommend for my dinner?



Thinking ...

Retrieved support set

Multi-memories supports. Denoted as I_q^* .

episodic

User is sick on September 26.

user_profile

User is sick and needs light food.

agent_method

Recommend light and no spicy food.

Valid entries

QA Answer



Here are some light foods for you: porridge and vegetable soup.

